



Node of Ranvier

Gap between segments of myelin sheath on an axon

Myelin sheath

Series of fatty wrappings along an axon; insulates axons to prevent short-circuiting and speeds up the transmission of nerve impulses

NEURONAL NETWORK

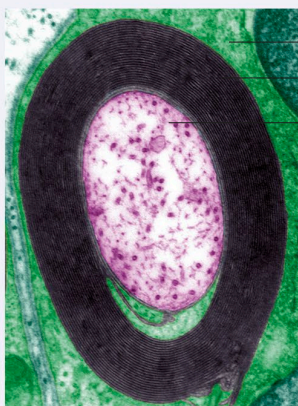
The snaking dendrites and axons of a neural net, which are reaching out to communicate, are clearly visible in this image. These neurons are of the multipolar type, which are found especially in the cortex of the brain. A single neuron can correspond via its processes with tens of thousands of others.

Synaptic knob

End of an axon fibre

MYELINATED NERVES

The axons of most nerve fibres are wrapped in thin layers of a white, fatty substance called myelin. These create a sheath of insulation that allows nerve impulses to be conducted quickly along the length of the nerve fibre.



Schwann cell

Myelin layer

Axon

SCHWANN CELL

The myelin sheath that insulates a nerve fibre in the peripheral nervous system is produced by a Schwann cell, which is wrapped spirally around the body of the axon.